

ORGANIC CHEMISTRY – ALCOHOLS

KEY VOCABULARY

Alcohol: organic molecule with the –OH functional group.

Ethanol: the alcohol in drinks, C₂H₅OH.

Fermentation: yeast turns sugar into ethanol + CO₂.

Functional group: the –OH group, where reactions occur.

COMMON MISCONCEPTIONS

~~Alcohols are hydrocarbons.~~

✓ They contain oxygen too (–OH group).

~~Fermentation needs high temperatures.~~

✓ Yeast works best around 30–40°C.

~~Ethanol does not burn.~~

✓ Ethanol is flammable and used as a fuel.

PROPERTIES OF ALCOHOLS

First four: methanol, ethanol, propanol, butanol.

Functional group is –OH.

Burn in air → CO₂ + H₂O.

React with sodium → hydrogen gas.

Oxidised to carboxylic acids.

1

FUNCTIONAL GROUP

Give the **functional group** of an alcohol. [1 mark]

2

FERMENTATION

Name the two products made when yeast ferments glucose. [2 marks]

3

CONDITIONS

State the conditions needed for fermentation and why. [2 marks]

4

COMBUSTION

Write the products of the complete combustion of ethanol. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Fermentation makes ____ and ____ gas.

ethanol

hydrogen

carbon dioxide

oxygen

6

USES

Give two uses of ethanol. [2 marks]